Kernix 0.12.0 / Performance brief

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

51 workloads

174 observations

Source precision retained

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

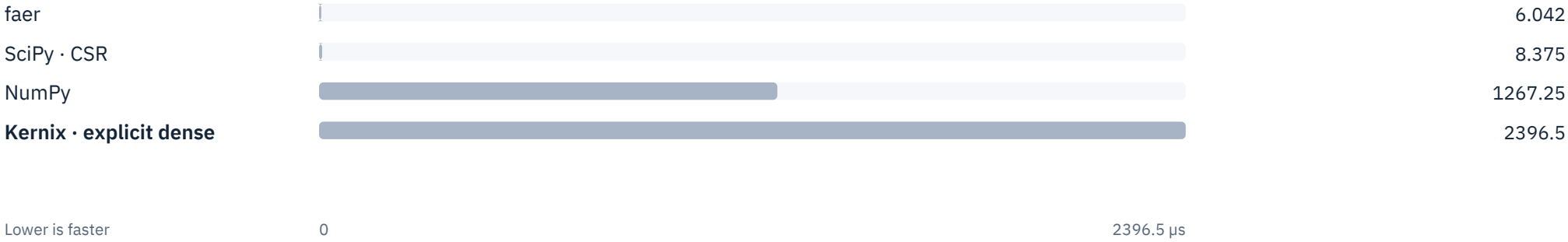
Kernix 0.12.0 / Sparse matrices & memory

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Sparse matrix × vector

Per-call median

2048 × 2048 · five diagonals · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Sparse × dense matrix

Per-call median

512 × 512 · 32 RHS · five diagonals · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Kernix 0.12.0 / Sparse matrices & memory

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Sparse × dense matrix

512 × 512 · 32 RHS · five diagonals · f32 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Sparse × sparse matrix

256 × 256 · five diagonals each · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



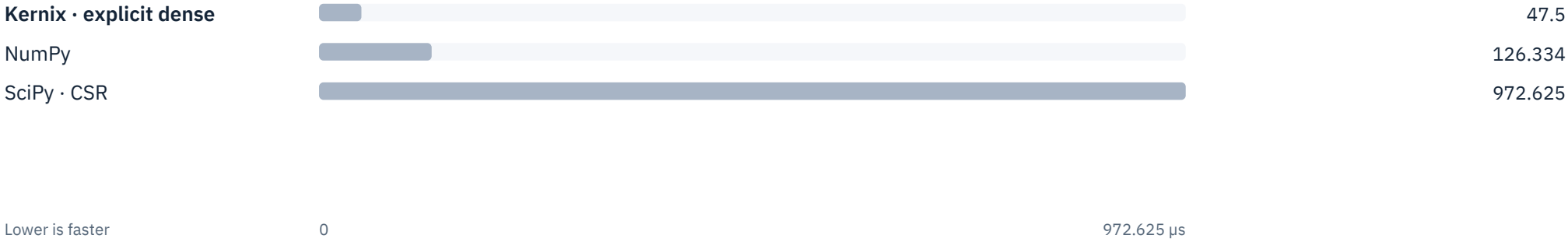
Kernix 0.12.0 / Sparse matrices & memory

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Automatic CSR construction

Per-call median

512 × 512 · clone + scan + compress · f64 · Construction · dense copy or compression · preparation costs included



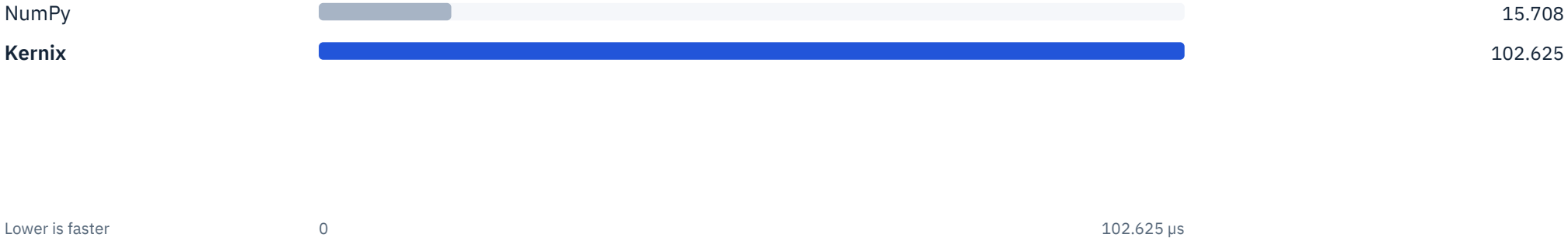
Kernix 0.12.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication · transposed view

Per-call median

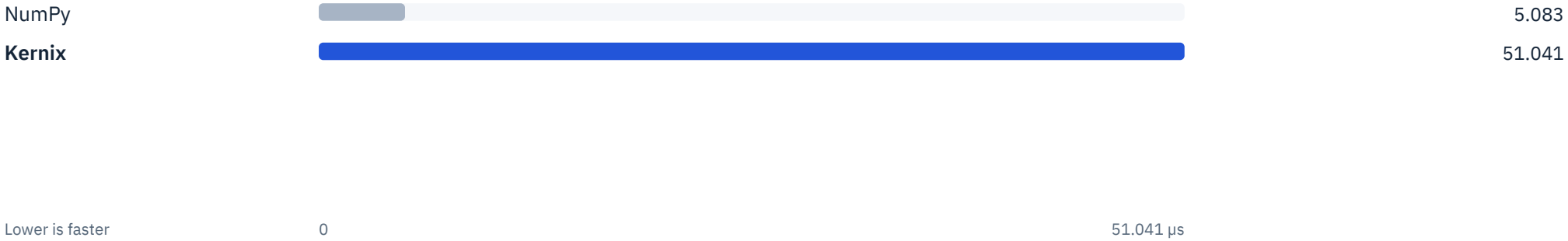
128 × 128 · 128 × 128 · f64 · 1 worker · borrowed transpose · allocated output · independent operands



Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · borrowed transpose · allocated output · independent operands



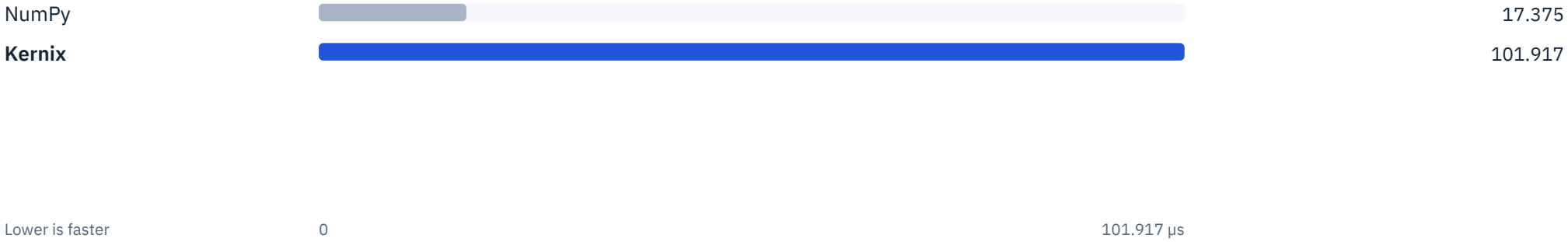
Kernix 0.12.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Independent matrix operands

Per-call median

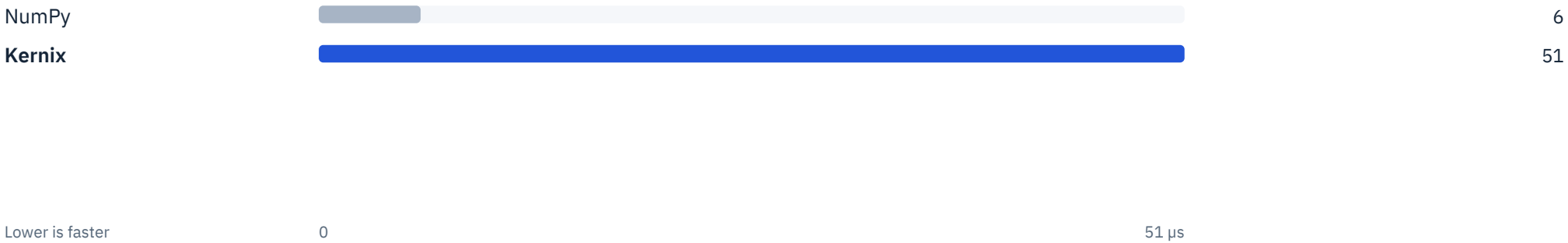
128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



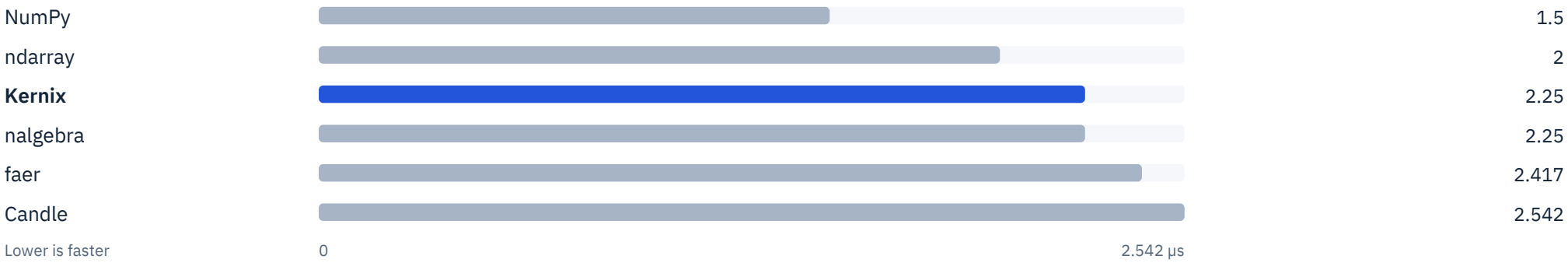
Kernix 0.12.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product

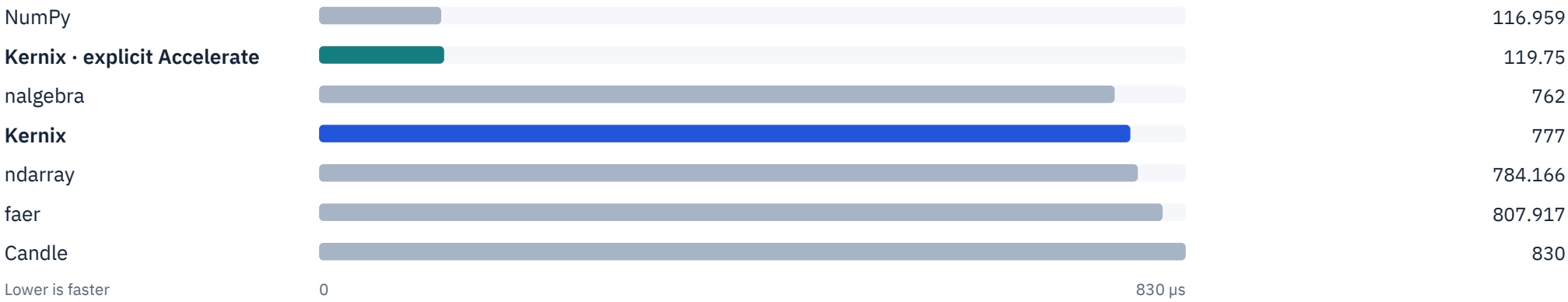


Kernix 0.12.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



Kernix 0.12.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



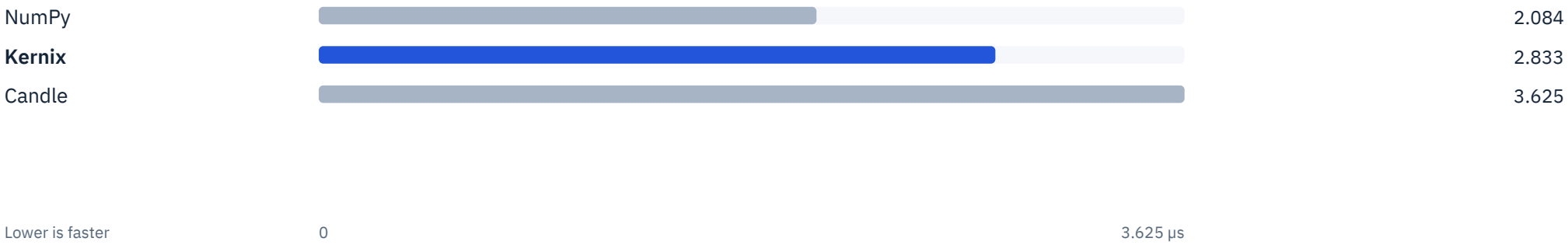
Kernix 0.12.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.12.0 / Explicit parallel execution

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.12.0 / Explicit parallel execution

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

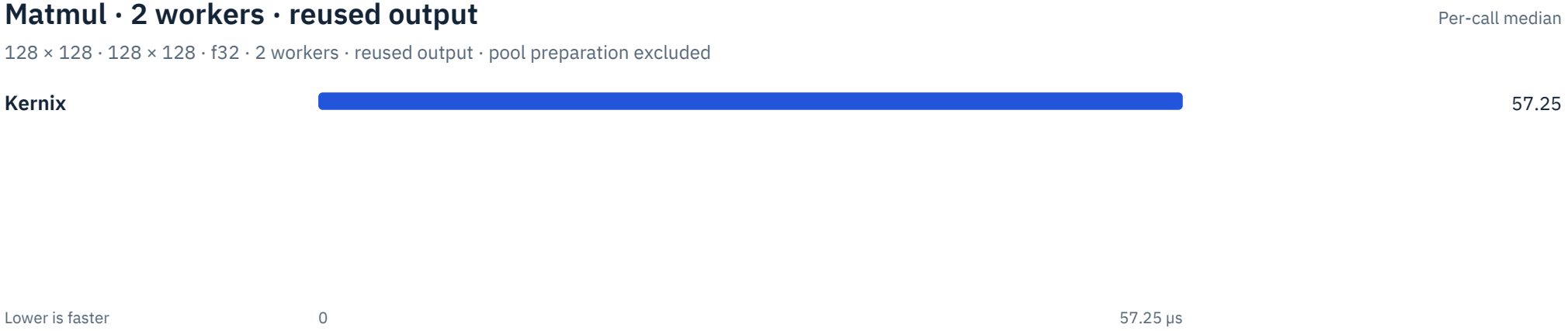
Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



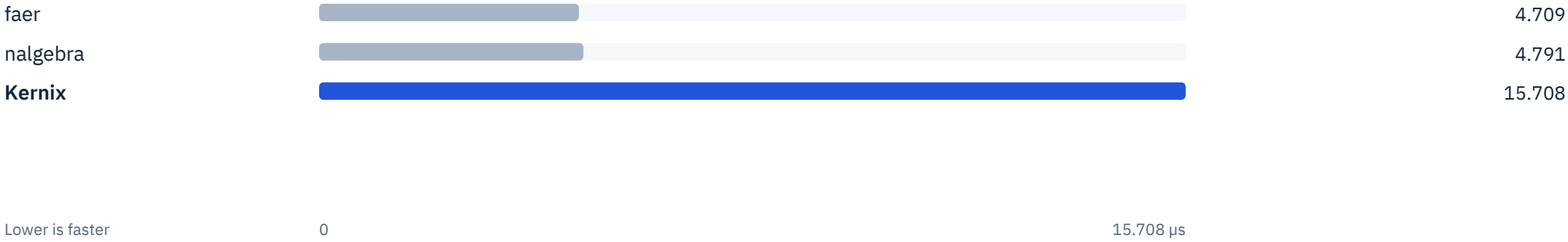
Kernix 0.12.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky solve · reused factor

Per-call median

128 × 128, 1 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Cholesky solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Kernix 0.12.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky solve · reused factor

Per-call median

512 × 512, 128 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



LU solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



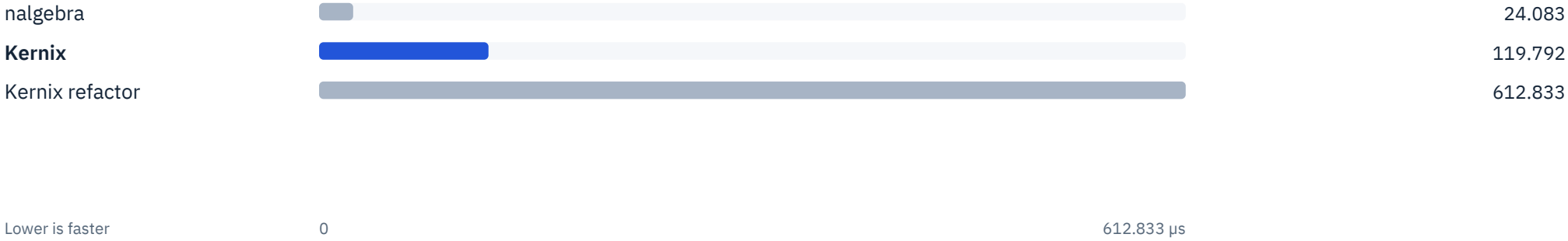
Kernix 0.12.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky rank-one update

Per-call median

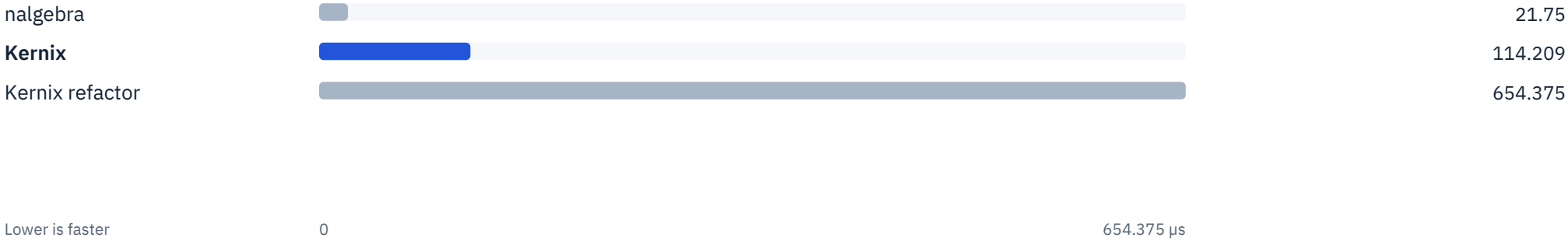
256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Cholesky rank-one downdate

Per-call median

256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



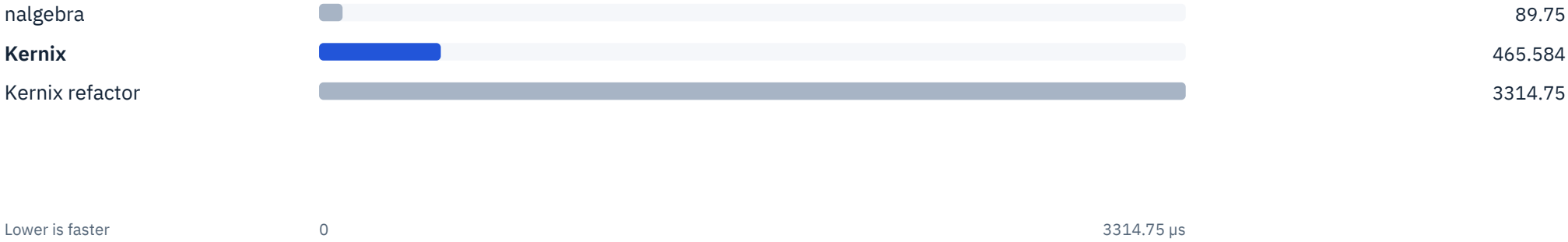
Kernix 0.12.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky rank-one update

512 × 512 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix

Per-call median



Thin QR

512 × 128 · f64 · 1 worker · allocated output

Per-call median



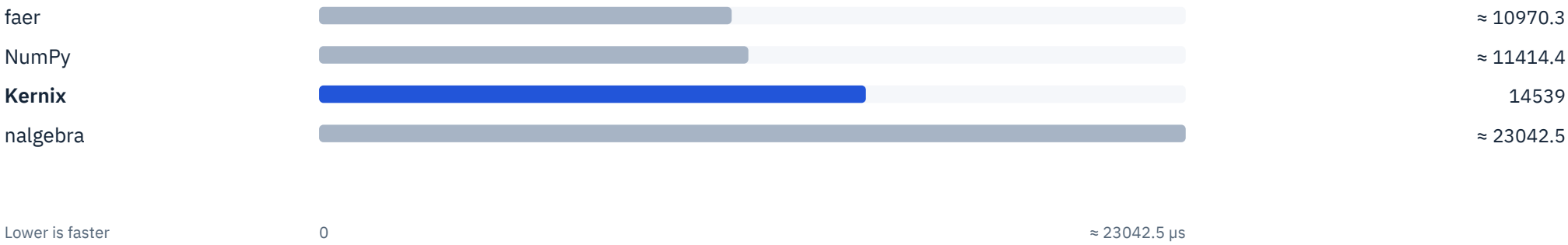
Kernix 0.12.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Thin QR

1024 × 256 · f64 · 1 worker · allocated output

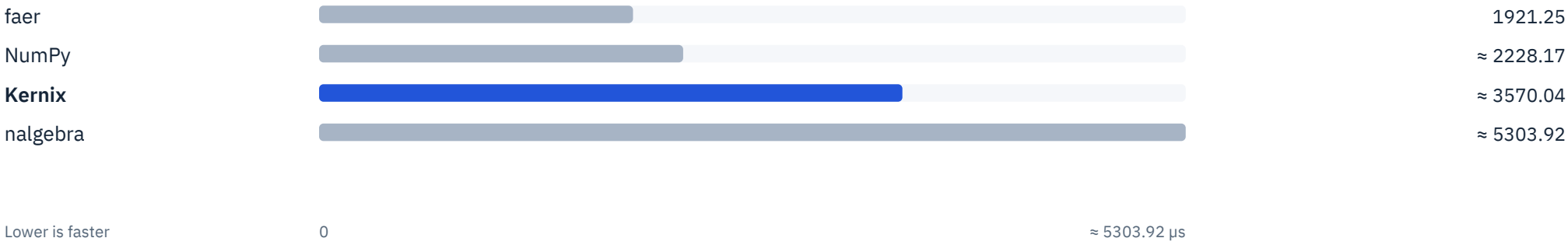
Per-call median



Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

Per-call median



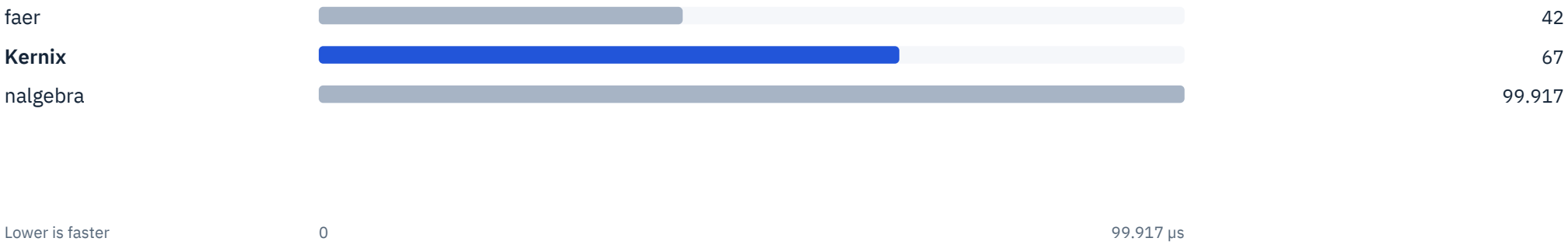
Kernix 0.12.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

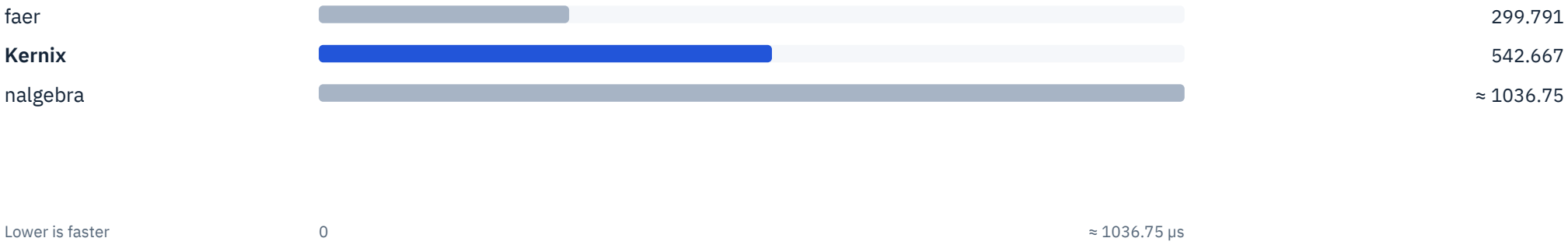
Per-call median



Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



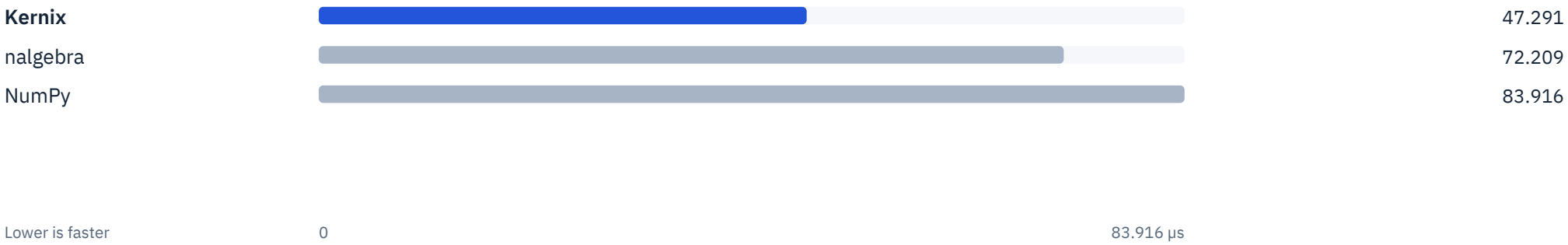
Kernix 0.12.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Minimum-norm least squares

Per-call median

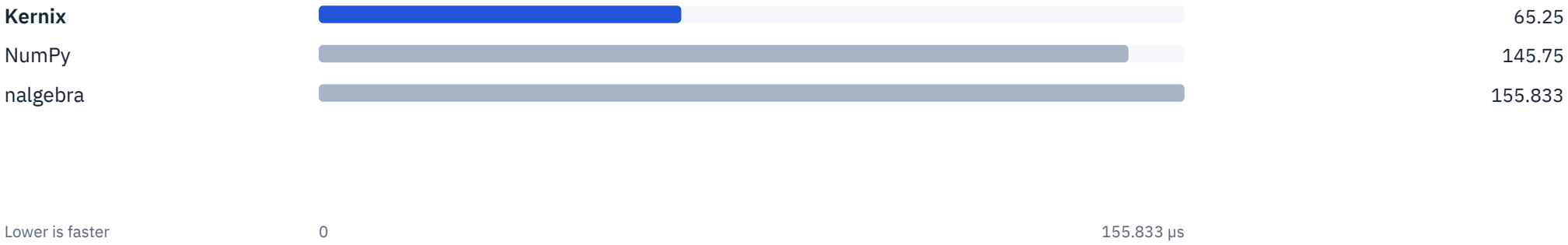
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Kernix 0.12.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



Kernix 0.12.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky factorization

Per-call median

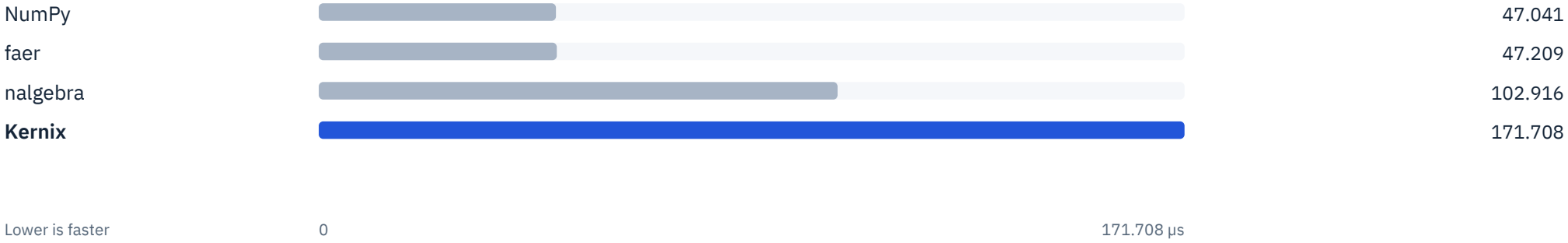
64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output

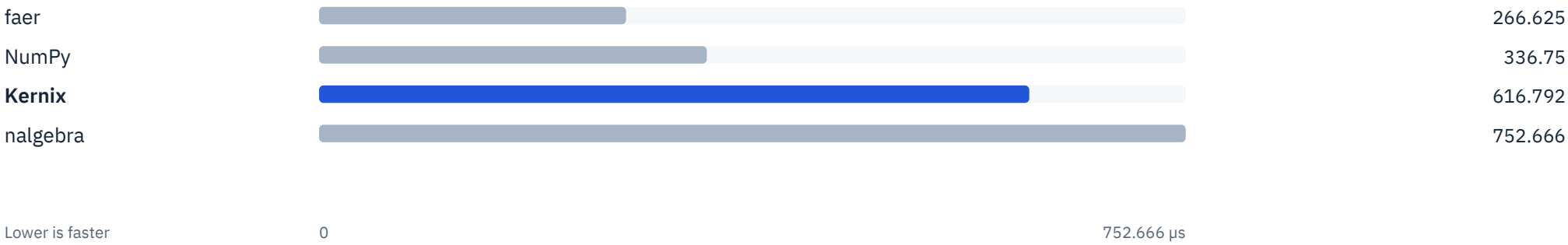


Kernix 0.12.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output



Kernix 0.12.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.12.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.12.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

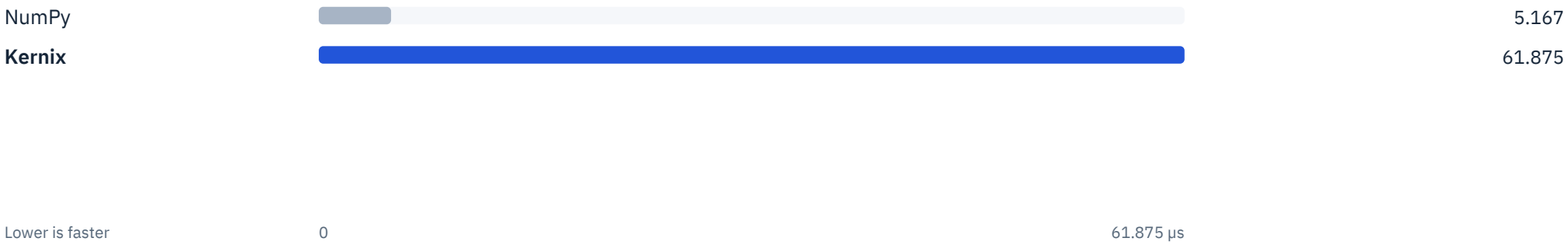


Kernix 0.12.0 / Tensor contractions

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output



Kernix 0.12.0 / Statistics & activations

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Row softmax

256 × 64 · f64 · 1 worker · allocated output



Kernix 0.12.0 / Tensor construction

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

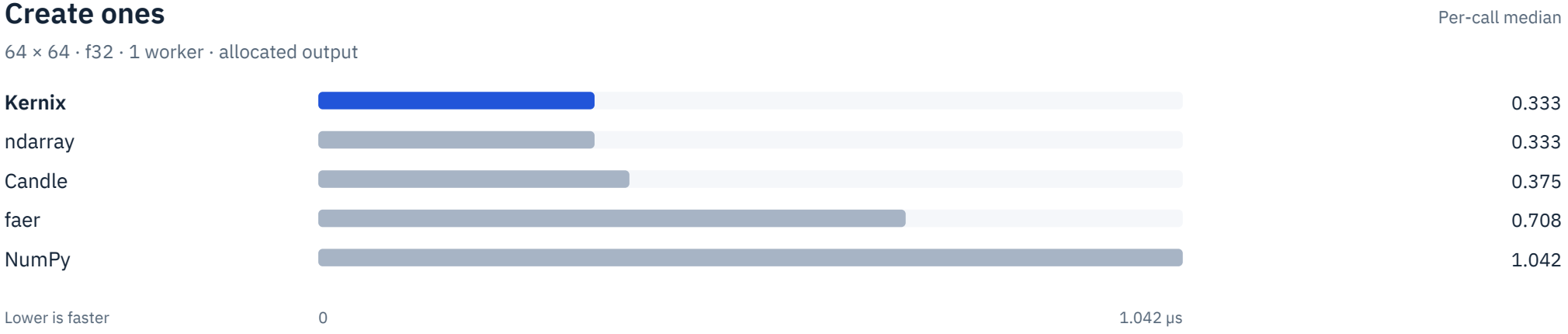
Create zeros

64 × 64 · f32 · 1 worker · allocated output



Create ones

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Sparse matrix × vector / SciPy · CSR 2048 × 2048 · five diagonals · f64	8375	8.375
Sparse matrix × vector / NumPy 2048 × 2048 · five diagonals · f64	1267250	1267.25
Sparse matrix × vector / faer 2048 × 2048 · five diagonals · f64	6042	6.042
Sparse matrix × vector / Kernix · explicit dense 2048 × 2048 · five diagonals · f64	2396500	2396.5
Sparse × dense matrix / SciPy · CSR 512 × 512 · 32 RHS · five diagonals · f64	49666	49.666
Sparse × dense matrix / Kernix · explicit dense 512 × 512 · 32 RHS · five diagonals · f64	438875	438.875
Sparse × dense matrix / NumPy 512 × 512 · 32 RHS · five diagonals · f64	86750	86.75
Sparse × dense matrix / faer 512 × 512 · 32 RHS · five diagonals · f64	53167	53.167
Sparse × dense matrix / SciPy · CSR 512 × 512 · 32 RHS · five diagonals · f32	13875	13.875
Sparse × dense matrix / Kernix · explicit dense 512 × 512 · 32 RHS · five diagonals · f32	237042	237.042

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Sparse × dense matrix / NumPy 512 × 512 · 32 RHS · five diagonals · f32	30458	30.458
Sparse × dense matrix / faer 512 × 512 · 32 RHS · five diagonals · f32	45958	45.958
Sparse × sparse matrix / SciPy · CSR 256 × 256 · five diagonals each · f64	43958	43.958
Sparse × sparse matrix / NumPy 256 × 256 · five diagonals each · f64	124500	124.5
Sparse × sparse matrix / Kernix · explicit dense 256 × 256 · five diagonals each · f64	820334	820.334
Automatic CSR construction / NumPy 512 × 512 · clone + scan + compress · f64	126334	126.334
Automatic CSR construction / SciPy · CSR 512 × 512 · clone + scan + compress · f64	972625	972.625
Automatic CSR construction / Kernix · explicit dense 512 × 512 · clone + scan + compress · f64	47500	47.5
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f64	15708	15.708
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f64	102625	102.625

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f32	51041	51.041
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f32	5083	5.083
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	17375	17.375
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	101917	101.917
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	51000	51
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	6000	6
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2250	2.25
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	1500	1.5
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	2000	2
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2417	2.417

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2542	2.542
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2250	2.25
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	16583	16.583
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	3250	3.25
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	14833	14.833
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	14333	14.333
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	14750	14.75
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	12792	12.792
Matrix multiplication / Kernix · explicit Accelerate 128 × 128 · 128 × 128 · f64	18250	18.25
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	108292	108.292

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	101458	101.458
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	118500	118.5
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	17875	17.875
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	107875	107.875
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	100500	100.5
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	807917	807.917
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	784166	784.166
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	830000	830
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	116959	116.959
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	777000	777

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	762000	762
Matrix multiplication / Kernix · explicit Accelerate 256 × 256 · 256 × 256 · f64	119750	119.75
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	8208	8.208
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	7958.000000000001	7.958000000000001
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	6667	6.667
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	7208	7.208
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	6917	6.917
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	1709	1.709
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	6000	6
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	51250	51.25

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	52792	52.792
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	52458	52.458
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	55292	55.292
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	51458	51.458
Matrix multiplication / Kernix · explicit Accelerate 128 × 128 · 128 × 128 · f32	5709	5.709
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	409917	409.917
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	389333	389.333
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	32208	32.208
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	426083	426.083
Matrix multiplication / Kernix · explicit Accelerate 256 × 256 · 256 × 256 · f32	35667	35.667

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	384209	384.209
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	370542	370.542
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	2084	2.084
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	3625	3.625
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	2833	2.833
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	2625	2.625
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	1834	1.834
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	2208	2.208
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	20917	20.917
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	125792	125.792

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	490000	490
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	57250	57.25
Cholesky solve · reused factor / nalgebra 128 × 128, 1 RHS · f64	4791	4.791
Cholesky solve · reused factor / Kernix 128 × 128, 1 RHS · f64	15708	15.708
Cholesky solve · reused factor / faer 128 × 128, 1 RHS · f64	4709	4.709
Cholesky solve · reused factor / faer 256 × 256, 64 RHS · f64	291083	291.083
Cholesky solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1343459	1343.459
Cholesky solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	1037832.9999999999	1037.8329999999999
Cholesky solve · reused factor / nalgebra 512 × 512, 128 RHS · f64	7494500	7494.5
Cholesky solve · reused factor / faer 512 × 512, 128 RHS · f64	2329209	2329.209

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky solve · reused factor / Kernix 512 × 512, 128 RHS · f64	9704709	9704.709
LU solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	943834	943.834
LU solve · reused factor / faer 256 × 256, 64 RHS · f64	272208	272.208
LU solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1148833	1148.833
Cholesky rank-one update / Kernix refactor 256 × 256 · f64	612833	612.833
Cholesky rank-one update / nalgebra 256 × 256 · f64	24083	24.083
Cholesky rank-one update / Kernix 256 × 256 · f64	119792	119.792
Cholesky rank-one downdate / Kernix refactor 256 × 256 · f64	654375	654.375
Cholesky rank-one downdate / nalgebra 256 × 256 · f64	21750	21.75
Cholesky rank-one downdate / Kernix 256 × 256 · f64	114209	114.209

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky rank-one update / Kernix refactor 512 × 512 · f64	3314750	3314.75
Cholesky rank-one update / Kernix 512 × 512 · f64	465584	465.584
Cholesky rank-one update / nalgebra 512 × 512 · f64	89750	89.75
Thin QR / Kernix 512 × 128 · f64	2620708	2620.708
Thin QR / NumPy 512 × 128 · f64	2729250	2729.25
Thin QR / faer 512 × 128 · f64	1698417	1698.417
Thin QR / nalgebra 512 × 128 · f64	3168125	3168.125
Thin QR / nalgebra 1024 × 256 · f64	23042542	23042.542
Thin QR / Kernix 1024 × 256 · f64	14539000	14539
Thin QR / faer 1024 × 256 · f64	10970291	10970.291

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / NumPy 1024 × 256 · f64	11414375	11414.375
Cholesky factorization / faer 512 × 512 · f64	1921250	1921.25
Cholesky factorization / NumPy 512 × 512 · f64	2228167	2228.167
Cholesky factorization / Kernix 512 × 512 · f64	3570042	3570.042
Cholesky factorization / nalgebra 512 × 512 · f64	5303917	5303.917
Column-pivoted QR / faer 96 × 32 · f64	42000	42
Column-pivoted QR / nalgebra 96 × 32 · f64	99917	99.917
Column-pivoted QR / Kernix 96 × 32 · f64	67000	67
Column-pivoted QR / nalgebra 256 × 64 · f64	1036749.9999999999	1036.7499999999999
Column-pivoted QR / faer 256 × 64 · f64	299791	299.791

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / Kernix 256 × 64 · f64	542667	542.667
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	83916	83.916
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	72209	72.209
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	47291	47.291
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	155833	155.833
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	145750	145.75
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	65250	65.25
Thin QR / nalgebra 96 × 32 · f64	48042	48.042
Thin QR / faer 96 × 32 · f64	45792	45.792
Thin QR / NumPy 96 × 32 · f64	58250	58.25

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / Kernix 96 × 32 · f64	58292	58.292
Thin QR / NumPy 256 × 64 · f64	395500	395.5
Thin QR / nalgebra 256 × 64 · f64	382583	382.583
Thin QR / Kernix 256 × 64 · f64	548750	548.75
Thin QR / faer 256 × 64 · f64	293625	293.625
Cholesky factorization / NumPy 64 × 64 · f64	15417	15.417
Cholesky factorization / Kernix 64 × 64 · f64	21917	21.917
Cholesky factorization / nalgebra 64 × 64 · f64	20750	20.75
Cholesky factorization / faer 64 × 64 · f64	10500	10.5
Cholesky factorization / nalgebra 128 × 128 · f64	102916	102.916

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / NumPy 128 × 128 · f64	47041	47.041
Cholesky factorization / faer 128 × 128 · f64	47209	47.209
Cholesky factorization / Kernix 128 × 128 · f64	171708	171.708
Cholesky factorization / faer 256 × 256 · f64	266625	266.625
Cholesky factorization / NumPy 256 × 256 · f64	336750	336.75
Cholesky factorization / nalgebra 256 × 256 · f64	752666	752.666
Cholesky factorization / Kernix 256 × 256 · f64	616792	616.792
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	6000	6
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	4125	4.125
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3250	3.25

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	6167	6.167
Model inference / Kernix batched normalized/sliced MLP · f64	2292	2.292
Model inference / Candle batched normalized/sliced MLP · f64	16959	16.959
Model recording + forward / Candle batched normalized/sliced MLP · f64	6250	6.25
Model recording + forward / Kernix batched normalized/sliced MLP · f64	3291	3.291
Model backward / Kernix prepared graph → fresh gradients · f64	3416	3.416
Model backward / Candle prepared graph → fresh gradients · f64	21708	21.708
Parameter update / Kernix transactional SGD, fixed gradients · f64	750	0.75
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	61875	61.875
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	5167	5.167

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Row softmax / Candle 256 × 64 · f64	82834	82.834
Row softmax / Kernix 256 × 64 · f64	83709	83.709
Row softmax / NumPy 256 × 64 · f64	76167	76.167
Row softmax / ndarray 256 × 64 · f64	61124.99999999999	61.12499999999999
Create zeros / ndarray 64 × 64 · f32	333	0.333
Create zeros / NumPy 64 × 64 · f32	500	0.5
Create zeros / faer 64 × 64 · f32	292	0.292
Create zeros / Candle 64 × 64 · f32	375	0.375
Create zeros / Kernix 64 × 64 · f32	375	0.375
Create ones / faer 64 × 64 · f32	708	0.708

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create ones / Candle 64 × 64 · f32	375	0.375
Create ones / ndarray 64 × 64 · f32	333	0.333
Create ones / Kernix 64 × 64 · f32	333	0.333
Create ones / NumPy 64 × 64 · f32	1042	1.042